

Titration report



Method

Method ID	CDC_M003	Release state	Released
Name	Titration with HCl - Liquid	Modified by	PredefinedUser
Modified on	2/11/2026 2:28:34 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.031512 mL
		Concentration R2	0.10063 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	4.996466 mL
		Concentration R2	0.09993 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.031680 mL
		Concentration R2	0.10063 mol/L

Statistics

R1

Mean	5.019886 mL	Minimum	4.996466 mL
Standard deviation	0.020282 mL	Rel. standard deviation	0.404 %
Total samples	3	Samples excluded	0

R2

Mean	0.10040 mol/L	Minimum	0.09993 mol/L
Standard deviation	0.00040 mol/L	Rel. standard deviation	0.403 %
Total samples	3	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.031512 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10063 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	4.996466 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.09993 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.031680 mL	OK

Titration report

LabX

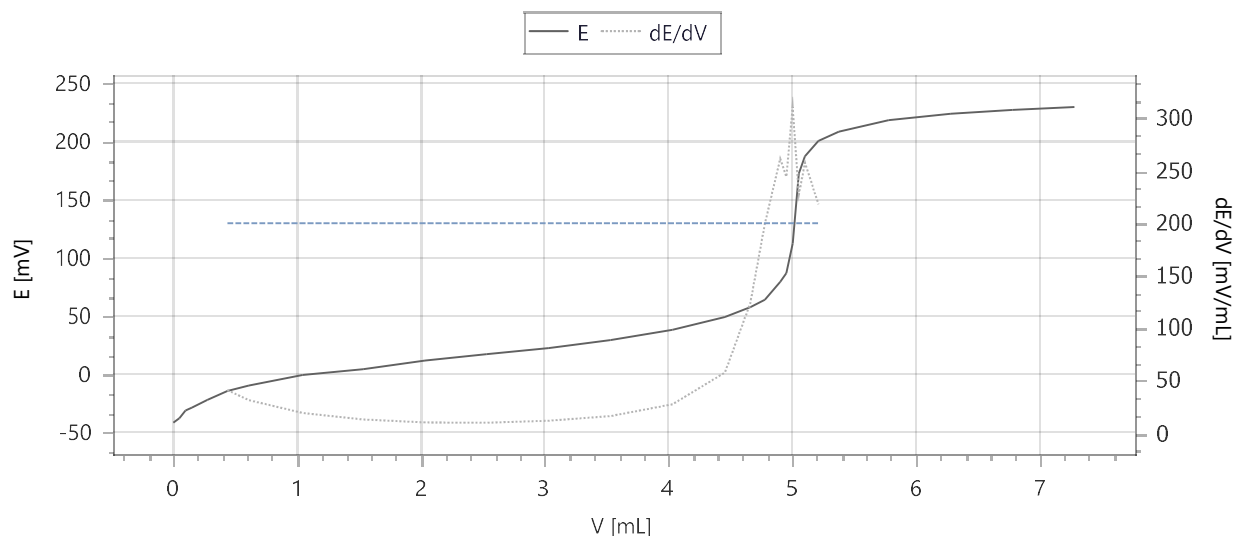
0.1 M TRIS - 5 mL
in 60 mL H₂O

Concentration R2

0.10063 mol/L OK

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3



Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-41.1		25.0			
3	0.0500	-37.2	0.0500	25.0	3.9		
7	0.1000	-30.6	0.0500	25.0	6.6		
10	0.1500	-28.0	0.0500	25.0	2.6		
14	0.2750	-21.6	0.1250	25.0	6.4		
17	0.4370	-13.8	0.1620	25.0	7.8	41.16	
20	0.6110	-9.2	0.1740	25.0	4.6	31.60	
24	1.0460	-0.1	0.4350	25.0	9.1	19.39	
29	1.5335	4.7	0.4875	25.0	4.8	13.19	
32	2.0335	12.1	0.5000	25.0	7.4	10.37	
35	2.5335	17.6	0.5000	25.0	5.5	10.08	
38	3.0335	22.8	0.5000	25.0	5.2	12.01	
41	3.5335	29.7	0.5000	25.0	6.9	16.45	
44	4.0335	38.4	0.5000	25.0	8.7	27.67	
48	4.4570	49.5	0.4235	25.0	11.1	57.89	
51	4.6620	58.1	0.2050	25.0	8.6	124.00	
54	4.7795	64.2	0.1175	25.0	6.1	199.49	
57	4.9035	79.6	0.1240	25.0	15.4	261.33	
61	4.9535	87.1	0.0500	25.0	7.5	244.29	
72	5.0035	112.6	0.0500	25.0	25.5	310.64	
	5.031512	146.2				315.47	EQP
79	5.0535	172.5	0.0500	25.0	59.9	227.19	
82	5.1035	187.3	0.0500	25.0	14.8	257.58	
85	5.2120	200.4	0.1085	25.0	13.1	217.98	
88	5.3760	208.4	0.1640	25.0	8.0		

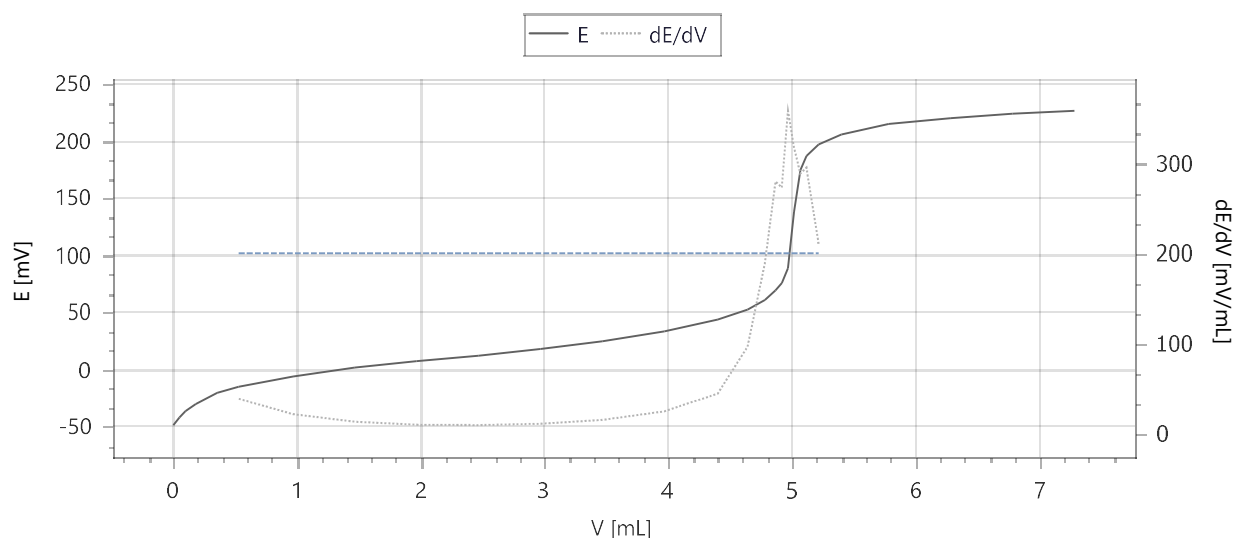
Titration report

LabX

91	5.7805	218.2	0.4045	25.0	9.8
94	6.2805	223.6	0.5000	25.0	5.4
97	6.7805	227.1	0.5000	25.0	3.5
100	7.2805	229.6	0.5000	25.0	2.5

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-48.2		25.0			
3	0.0500	-41.6	0.0500	25.0	6.6		
6	0.1000	-36.0	0.0500	25.0	5.6		
10	0.1830	-29.9	0.0830	25.0	6.1		
14	0.3550	-20.1	0.1720	25.0	9.8		
18	0.5300	-14.8	0.1750	25.0	5.3	38.90	
23	0.9675	-5.7	0.4375	25.0	9.1	21.96	
26	1.4675	2.0	0.5000	25.0	7.7	13.53	
30	1.9675	7.6	0.5000	25.0	5.6	10.22	
34	2.4675	12.6	0.5000	25.0	5.0	9.80	
37	2.9675	18.4	0.5000	25.0	5.8	11.30	
40	3.4675	25.1	0.5000	25.0	6.7	15.56	
43	3.9675	33.7	0.5000	25.0	8.6	25.01	
46	4.3995	44.0	0.4320	25.0	10.3	44.68	
49	4.6405	52.9	0.2410	25.0	8.9	96.53	
52	4.7800	61.1	0.1395	25.0	8.2	188.51	
56	4.8670	69.5	0.0870	25.0	8.4	279.51	
59	4.9170	75.9	0.0500	25.0	6.4	272.66	
62	4.9670	89.0	0.0500	25.0	13.1	357.55	
	4.996466	117.9				357.55	EQP
76	5.0170	138.0	0.0500	25.0	49.0	316.73	
80	5.0670	174.9	0.0500	25.0	36.9	288.13	
83	5.1170	187.2	0.0500	25.0	12.3	294.79	

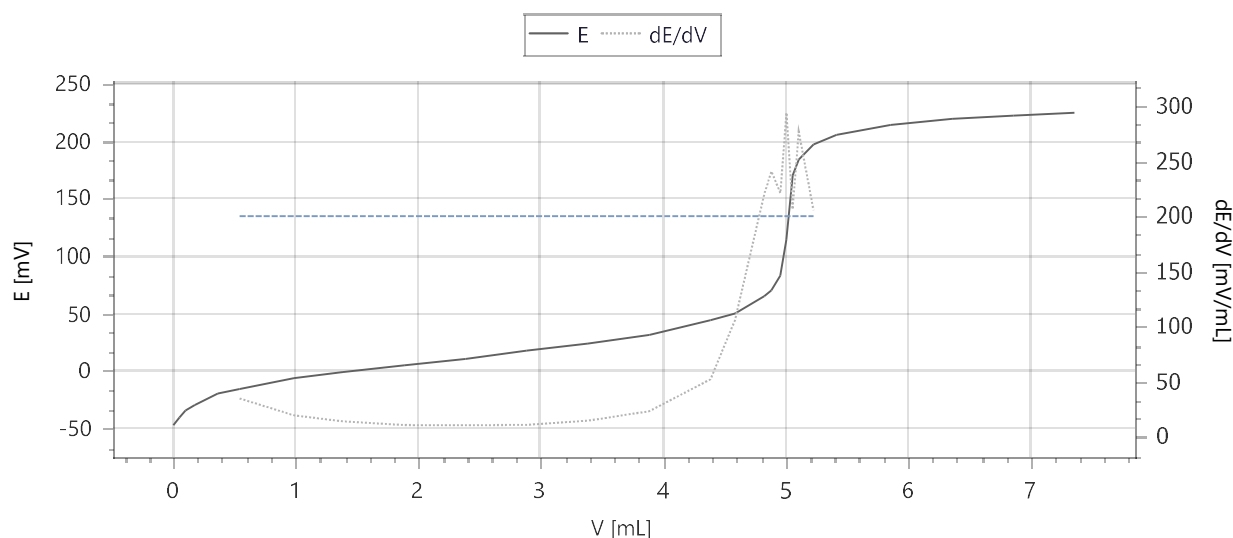
Titration report

LabX

86	5.2145	197.3	0.0975	25.0	10.1	210.16
89	5.3985	205.9	0.1840	25.0	8.6	
92	5.7825	215.1	0.3840	25.0	9.2	
95	6.2825	220.2	0.5000	25.0	5.1	
98	6.7825	224.0	0.5000	25.0	3.8	
101	7.2825	226.5	0.5000	25.0	2.5	

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-47.1		25.0			
3	0.0500	-40.2	0.0500	25.0	6.9		
6	0.1000	-34.1	0.0500	25.0	6.1		
10	0.1755	-29.4	0.0755	25.0	4.7		
14	0.3645	-19.4	0.1890	25.0	10.0		
19	0.5425	-15.5	0.1780	25.0	3.9	35.24	
25	0.9875	-6.2	0.4450	25.0	9.3	19.93	
30	1.3865	-0.9	0.3990	25.0	5.3	14.55	
34	1.8865	5.1	0.5000	25.0	6.0	11.21	
38	2.3865	10.5	0.5000	25.0	5.4	10.99	
41	2.8865	17.9	0.5000	25.0	7.4	11.40	
45	3.3865	23.9	0.5000	25.0	6.0	15.17	
48	3.8865	31.6	0.5000	25.0	7.7	23.72	
51	4.3865	44.3	0.5000	25.0	12.7	52.53	
54	4.5865	50.1	0.2000	25.0	5.8	106.03	
57	4.8255	65.2	0.2390	25.0	15.1	219.87	
60	4.8840	70.1	0.0585	25.0	4.9	240.62	
63	4.9570	82.9	0.0730	25.0	12.8	220.88	
74	5.0070	114.2	0.0500	25.0	31.3	293.52	
	5.031680	141.9				297.53	EQP
80	5.0570	170.4	0.0500	25.0	56.2	206.23	

Titration report

LabX

83	5.1070	184.1	0.0500	25.0	13.7	276.43
86	5.2275	197.1	0.1205	25.0	13.0	205.45
90	5.4150	205.4	0.1875	25.0	8.3	
93	5.8585	214.2	0.4435	25.0	8.8	
96	6.3585	219.3	0.5000	25.0	5.1	
99	6.8585	222.3	0.5000	25.0	3.0	
102	7.3585	224.7	0.5000	25.0	2.4	

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
11.02.2026 14:31:17	PredefinedUser	Administrator